Mediapipe Gesture Follow (No-PTZ)

1. Experiment Objective

Use MediaPipe gesture recognition to track the palm while also recognizing the gesture type, with body rotation tracking + distance following keeping the hand centered in the frame and maintaining a constant distance.

2. Experiment Principle

1. Key Terms

Term	Description
Gesture Recognition	MediaPipe Hands 21 keypoints → `fingersUp()` computes the states of the 5 fingers → classifies 6 gestures
Distance Follow	Estimates distance from the hand bounding box size, with PD control maintaining the desired distance
Fingertip Counting	Determines whether a finger is extended from the distance between the fingertip keypoints (4/8/12/16/20) and the corresponding knuckle
QoS	best_effort (image input + cmd_vel downlink), depth=1

2. Technical Architecture

```
/camera/image_raw (BEST_EFFORT)
  → gesture_tracker_node (enable_distance_follow:=true)
    ├── MediaPipe Hands → 21 keypoints → fingersUp() classification
    └── PD angular velocity + bounding box size distance following → /cmd_vel
```

> A MediaPipe Python environment is required (for both ROS2 and the workspace environment).

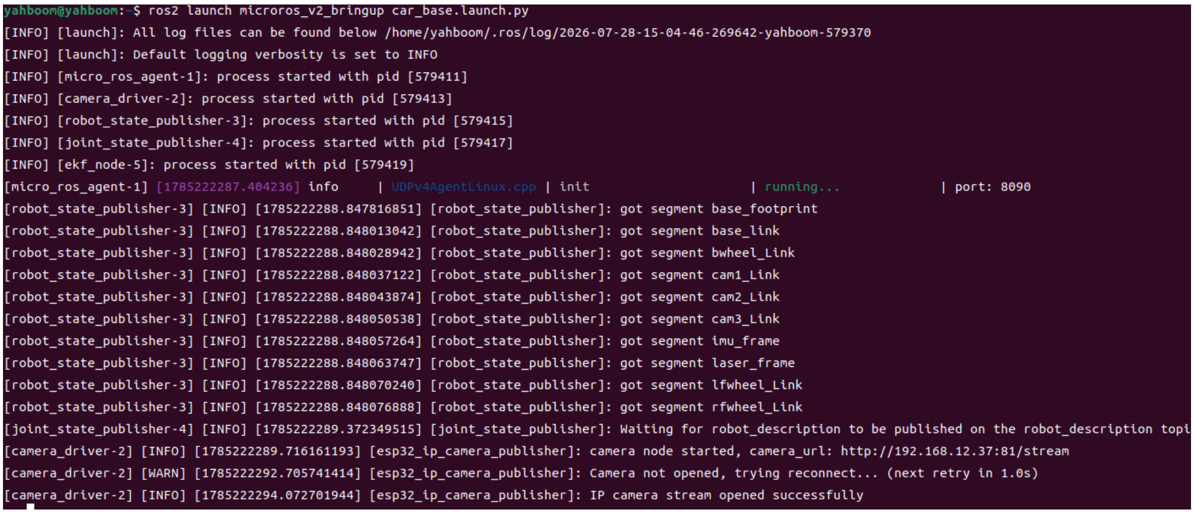
3. Experiment Steps

1. Hardware Preparation

Hardware	Requirement
PTZ Version	❌
No-PTZ Version	☑
Required Peripherals	ESP32 camera (mounted on bracket)

2. Start Agent + Camera + Joint Model (Terminal 1)

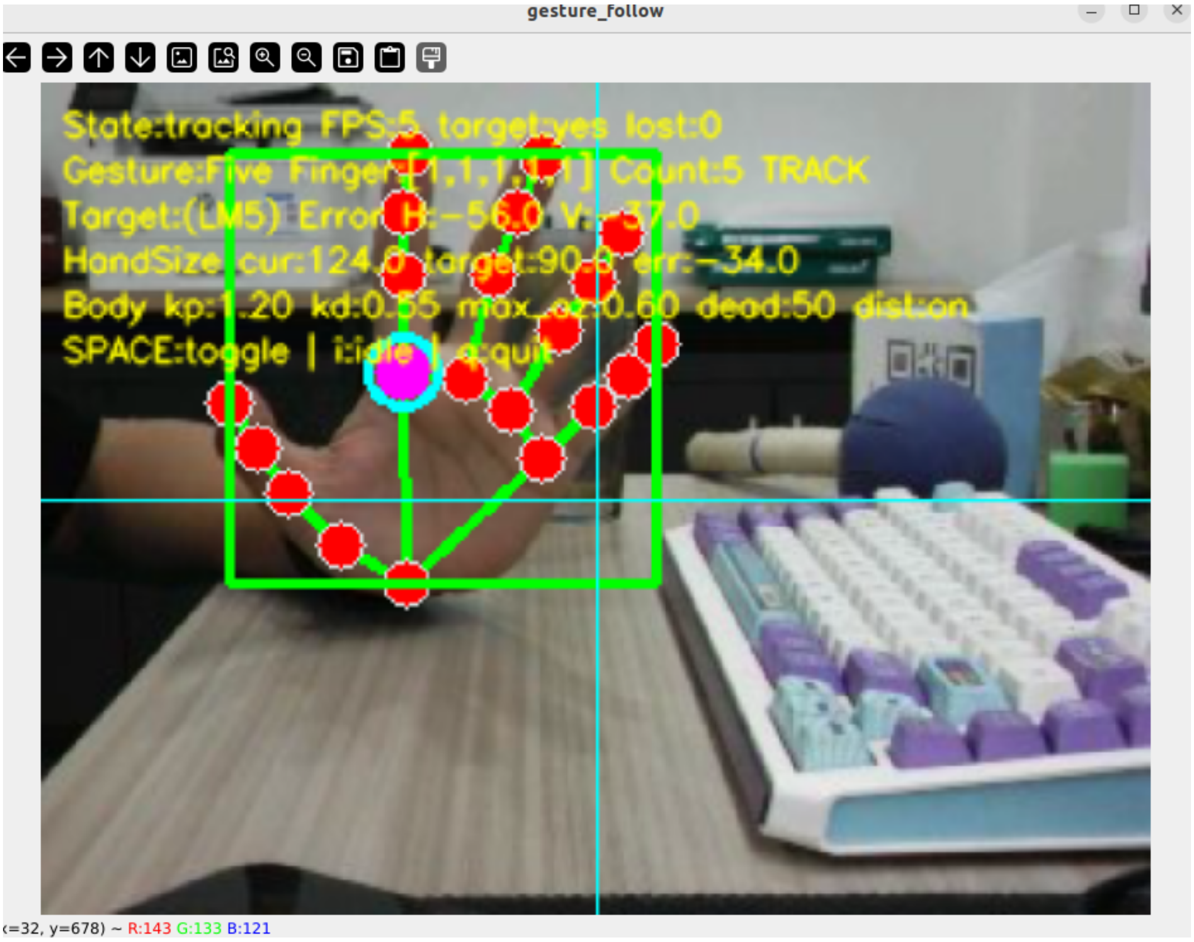
```
ros2 launch microros_v2_bringup car_base.launch.py
```



```
yahboom@yahboom:~$ ros2 launch microros_v2_bringup car_base.launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-07-28-15-04-46-269642-yahboom-579370
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [micro_ros_agent-1]: process started with pid [579411]
[INFO] [camera_driver-2]: process started with pid [579413]
[INFO] [robot_state_publisher-3]: process started with pid [579415]
[INFO] [joint_state_publisher-4]: process started with pid [579417]
[INFO] [ekf_node-5]: process started with pid [579419]
[micro_ros_agent-1] [1785222287.404236] info     | UDPv4AgentLinux.cpp | init                     | running...             | port: 8090
[robot_state_publisher-3] [INFO] [1785222288.847816851] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-3] [INFO] [1785222288.848013042] [robot_state_publisher]: got segment base_link
[robot_state_publisher-3] [INFO] [1785222288.848028942] [robot_state_publisher]: got segment bwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848037122] [robot_state_publisher]: got segment cam1_Link
[robot_state_publisher-3] [INFO] [1785222288.848043874] [robot_state_publisher]: got segment cam2_Link
[robot_state_publisher-3] [INFO] [1785222288.848050538] [robot_state_publisher]: got segment cam3_Link
[robot_state_publisher-3] [INFO] [1785222288.848057264] [robot_state_publisher]: got segment imu_frame
[robot_state_publisher-3] [INFO] [1785222288.848063747] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-3] [INFO] [1785222288.848070240] [robot_state_publisher]: got segment lfwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848076888] [robot_state_publisher]: got segment rfwheel_Link
[joint_state_publisher-4] [INFO] [1785222289.372349515] [joint_state_publisher]: Waiting for robot_description to be published on the robot_description topi
[camera_driver-2] [INFO] [1785222289.716161193] [esp32_ip_camera_publisher]: camera node started, camera_url: http://192.168.12.37:81/stream
[camera_driver-2] [WARN] [1785222292.705741414] [esp32_ip_camera_publisher]: Camera not opened, trying reconnect... (next retry in 1.0s)
[camera_driver-2] [INFO] [1785222294.072701944] [esp32_ip_camera_publisher]: IP camera stream opened successfully
```

3. Start Gesture Follow (Terminal 2)

```
ros2 launch microros_v2_no_ptz_visual_control_pkg gesture_follow.launch.py
```



6. Operation Instructions

Key	Function
Space	Toggle follow/standby
q / ESC	Exit

4. Experiment Result

- Hand approaches → the car moves backward; moves away → the car moves forward to follow
- The frame displays the gesture type in real time (5/4/3/2/1 fingers / fist)
- 21 keypoints + skeleton connections visualized
- Hand moves out of the frame → stops moving

5. Code Explanation

> Source code paths:
>
> - Launch: `~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/launch/gesture_follow.launch.py`
> - Node source: `~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/microros_v2_no_ptz_visual_control_pkg/gesture_tracker_node.py`
> - Hand detector: `~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/microros_v2_no_ptz_visual_control_pkg/hand_detector.py`

1. Core Node Explanation

`gesture_follow.launch.py` launches `gesture_tracker_node`, which adds `fingersUp()` gesture classification on top of palm tracking. Follow mode enables both rotation tracking and distance following — the former is based on the palm center x-coordinate error, and the latter on the bounding box size error.

2. Key Parameters

> Parameter definition files:
>
> - Launch: `~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/launch/gesture_follow.launch.py`
> - Node: `~/microros_ws/src/microros_v2_no_ptz_visual_control_pkg/microros_v2_no_ptz_visual_control_pkg/gesture_tracker_node.py`

Parameter	Type	Default	Description
`hand_detection_confidence`	float	`0.8`	Hand detection confidence threshold
`deadband_x_pixel`	float	`50.0` px	Horizontal deadband
`angular_kp` / `angular_kd`	float	`1.2` / `0.55`	Angle PD gains
`max_angular_z`	float	`0.6` rad/s	Maximum turning speed
`enable_distance_follow`	bool	`true`	Enable distance following
`target_hand_size_px`	float	`90.0` px	Desired hand size
`distance_tolerance_hand_size_px`	float	`12.0` px	Distance tolerance
`distance_kp` / `distance_kd`	float	`0.05` / `0.03`	Distance PD gains
`max_linear_x`	float	`0.2` m/s	Maximum forward/backward speed

3. Node Description

Node	Package	Description
`gesture_tracker_node`	`microros_v2_no_ptz_visual_control_pkg`	MediaPipe gesture tracking + following, finger classification + dual PD

6. Common Issues

Issue	Cause	Solution
Gesture not recognized	Insufficient lighting or hand too far	Improve lighting and move closer to the camera
Incorrect gesture classification	Insufficient fingertip detection accuracy	Keep fingers separated and clearly visible
Distance following overshoot	PD gains too large	Lower `distance_kp`